

Gauge and Homological Structures in Quantum Error Correction



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Talk plan

- Preliminary
- Logical gates by gauge field formalism
- Homological origin of transversality
- Summary

What is Quantum Error Correction?

- Quantum information is embedded into a **larger Hilbert space**
- Physical n qubit Hilbert space : $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$
- Stabilizer code : code states are defined as +1 simultaneous eigenstates of commutative Pauli operators (“stabilizer generator”)
- CSS code : X-type of Z-type stabilizers

$$S_i^X = \prod_{j=1}^n (X_j)^{(H_X)_{ij}}, \quad S_i^Z = \prod_{j=1}^n (Z_j)^{(H_Z)_{ij}} \quad \text{with } H_X H_Z^T = 0$$

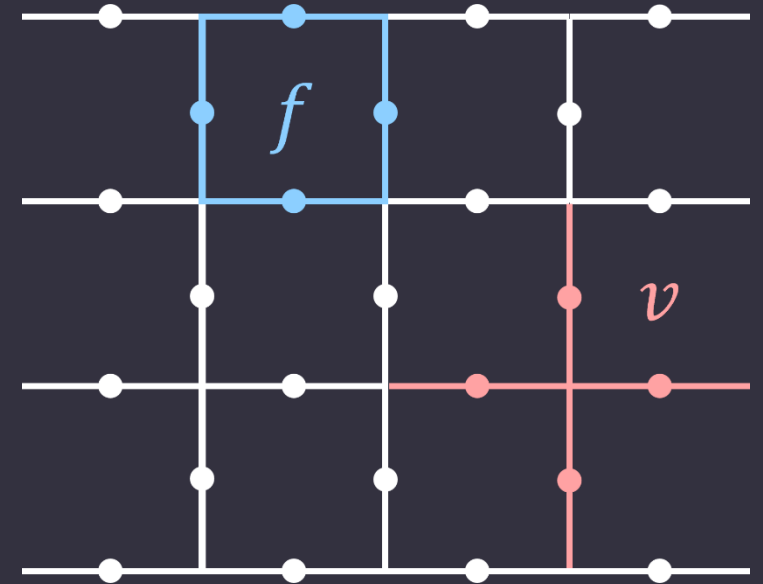
- Code distance d : Minimum weight of logical Pauli operators
- For high-energy physicists:
 - (X-)stabilizers = **gauge transformation** (Gauss law) generator
 - Code space = **gauge-invariant** subspace

Logical gates and universal computation

- Implementation of **logical gates beyond Pauli operator**
⇒ central problem in fault-tolerant quantum computation
- Logical Pauli op. $\bar{X}, \bar{Z}$: logical qubit operation while preserving $\mathcal{H}_c$
- Clifford hierarchy
 - Level 1: Pauli (X, Z)
 - Level 2: Clifford ($S, H, CNOT$)
 - Level m : $\exp(i\pi Z/2^{m-1}), C^{m-1}Z, \dots$ etc. ($m = 3; T$)
- Eastin-Knill theorem: No universal gate set only with transversal gates
- Challenge: **which** CSS code can implement **what kind** of transversal gates?

Toric code and gauge theory

- Toric code Hamiltonian = (topological) $\mathbb{Z}_2$ lattice gauge theory
- On 2D torus
 - vertex op. $A_v = \prod_{e \ni v} X_e \leftrightarrow X$ stab.
 - plaquette op. $B_p = \prod_{e \in \partial p} Z_e \leftrightarrow Z$ stab.
- Toric code Hamiltonian: $H = -\sum_v A_v - \sum_p B_p$
- Kogut-Susskind Hamiltonian
 - Z_e : gauge field on a link
 - A_v : Gauss law op.
 - B_p : magnetic flux (curvature)
- Degeneracy of ground states = 2^2
→ $k = 2$ logical qubits



CSS-homology correspondence

- CSS code is a length-2 chain complex

$$C_{\text{CSS}}: C_2 \xrightarrow{\partial_2 = H_Z^T} C_1 \xrightarrow{\partial_1 = H_X} C_0$$

C_1 (1-chain)	Physical qubits
C_2 (2-chain)	Z-stabilizers
C_0 (0-chain)	X-stabilizers
$H_1 := \ker \partial_1 / \text{im } \partial_2$ (homology group)	Logical Z operator
$H^1 := \ker \partial_2^T / \text{im } \partial_1^T$ (cohomology group)	Logical X operator
code distance	(co)systole

- For toric code,
 - chain complex = cell decomposition of torus
 - logical operator = non-trivial cycle (Wilson loop)

Recent developments

- Algebraic operation of chain complexes gives new family of QEC codes
- Examples
 - Hypergraph product codes → tensor product [Tillich and G. Zemor 2014]
 - Fiber-bundle codes → fiber bundle [Hastings, Haah, O' Donnell 2021]
 - Lifted product codes → lifted product [Panteleev, Kalachev 2021]
- Inclusion of measurement error leads to longer chain complexes
- Examples
 - Higher-dimensional QEC codes
 - Fault-complex [Hillmann et.al. 2025]
 - Single-shot QEC [Campbell 2018]
- Mathematical tools are useful to analyze these codes

Gauge field formalism

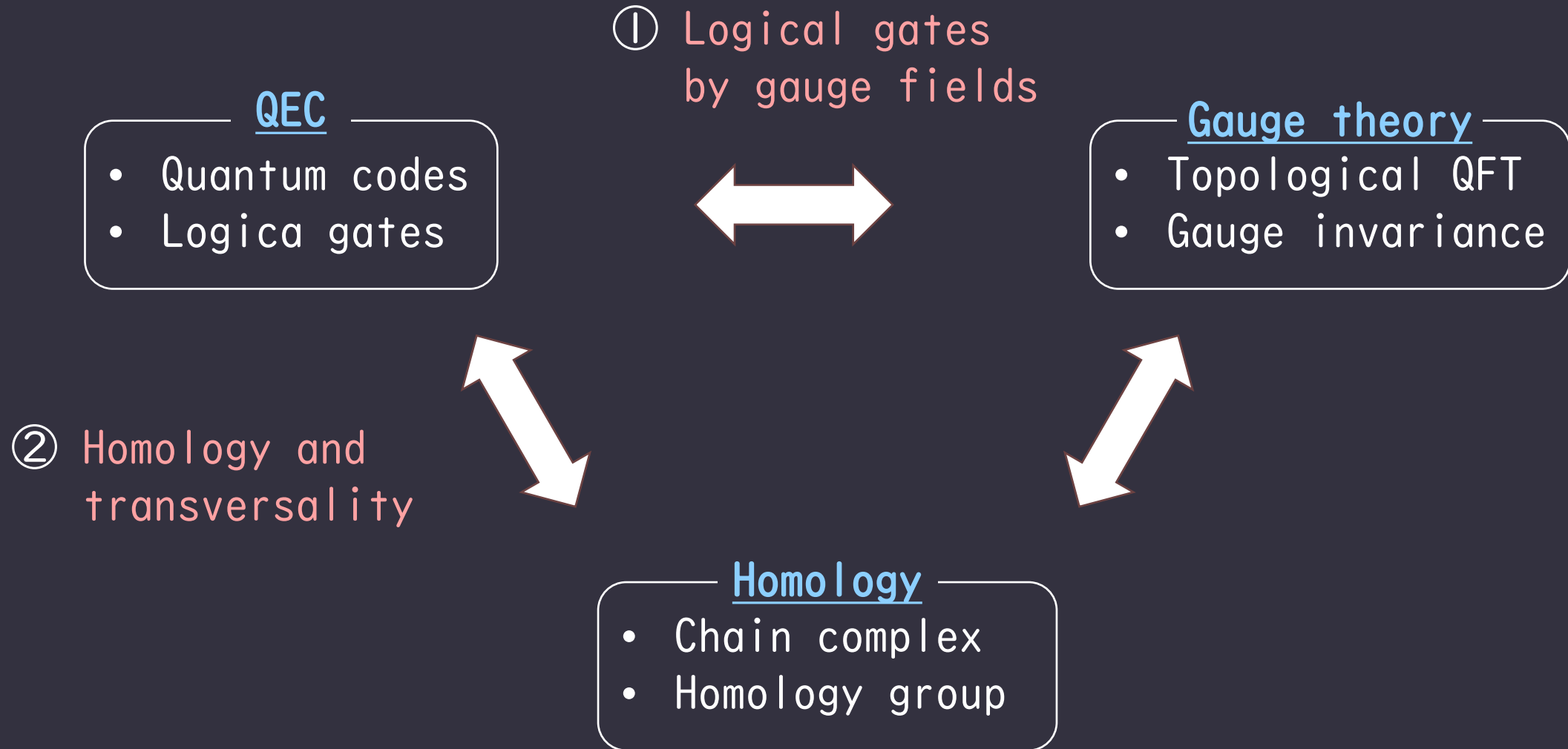
- Operator-valued cochains on chain complexes
= gauge field in lattice gauge theory
- Electric/magnetic field a, b : linear maps from 1-chain to operator

$$a(e_j) := z_j = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}_j, \quad b(e_j) = x_j = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}_j$$

$$\Rightarrow Z_j = \exp(i\pi a(e_j)), \quad X_j = \exp(i\pi b(e_j))$$

- Stabilizers : $S_i^Z = \exp(i\pi a(\partial_2 f_i))$, $S_i^X = \exp(i\pi b(\partial_1^T v_i))$
- Logical Z operator : $\bar{Z}(\gamma) = \exp(i\pi a(\gamma))$ with $\gamma \in H_1$
- Correspondence to lattice gauge theory
 - a = electric $\mathbb{Z}_2$ gauge field
 - $\bar{Z}(\gamma) = \text{Wilson loop } W(\gamma) = \exp(i \oint_\gamma a)$

Gauge-QEC-Homology correspondence?



Logical gates by gauge field formalism

Junichi Haruna
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Gauge field construction of logical gates

- Idea: express logical gates as exponential of polynomials of gauge fields

Logical Gate	Gauge field expression
$\bar{S}(\gamma)$	$\exp(i\pi a(\gamma)^2/2)$
$\bar{H}(\gamma)$	$\bar{S}(\gamma) \exp(i\pi b(\tilde{\gamma})^2/2) \bar{S}(\gamma)$
$\overline{CZ}(\gamma_1, \gamma_2)$	$\exp(i\pi a(\gamma_1) a(\gamma_2))$
$\bar{T}(\gamma)$	$\exp(i\pi(2a^3 - 3a^2 + 2a)/4)$

- No special structure on QEC codes: applicable for **general** CSS codes

Physical gate decomposition

- Explicit **physical gate decomposition** from gauge field expression
- Example: logical S gate

$$\bar{S}(\gamma) = \prod_{i=1}^n S_i \cdot \prod_{i < j} CZ_{i,j}$$

(product of S gate on all qubits on support of γ , and CZ gate of every pair of them)

- Notice: these gates are **NOT** necessarily fault-tolerant

Cohomology dependence

- The logical action of these gates depends only on the **cohomology class** of the logical qubits

$$U(\gamma) \simeq U(\gamma + \partial_2 f) \text{ on } \mathcal{H}_c$$

- Gauge transformation: $a(\gamma) \rightarrow a(\gamma) + dv$
 $\Rightarrow$ **Logical** operator = **Gauge-invariant** operator
- Open question
 - incorporating fault-tolerance or circuit-depth optimality
 - higher-order topological invariant
 - higher-form gauge fields

Homological origin of transversal implementability

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Classification of transversal gates

- Transversal Pauli Z rotation (Clifford level m)

$$U(\theta/2^{m-1}) := \prod_{j=1}^n \exp\left(i\pi \frac{\theta_j}{2^{m-1}} \frac{I - Z_j}{2}\right) \text{ with } \theta \in \mathbb{Z}_{2^m}^n$$

(e.g. $U(\theta/2) = \prod_i S_i^{\theta_i}$, $U(\theta/4) = \prod_i T_i^{\theta_i}$)

- Main theorem I

Logical action of $U(\theta/2^{m-1})$ is classified by a $\mathbb{Z}_{2^m}$ -module

$$H^1 \otimes_H S_m$$

($S_m := \bigoplus_{v=1}^m 2^{v-1}(\ker H_Z)^{\otimes_H (v-1)}$, $\otimes_H$: Hadamard product)

- Point: Extension of logical Pauli operators to higher Clifford levels with $\mathbb{Z}_{2^m}$ coefficient (rather than $\mathbb{Z}_2$)

Transversal implementability

- Main theorem 2
 $\pi/2^m$ rotation of logical Z operator can be implemented transversally if and only if a sequence $\{\theta_v \in \mathbb{Z}_{2^v}\} (v = 1, \dots, m)$ s.t.

$$\beta_1^{(v)}([\theta_v]) = \beta_2^{(v)}([\theta_v]) = 0$$

- $\beta_1^{(v)}$... compatibility obstruction at level m
 $\beta_2^{(v)}$... lifting obstruction from level m to $m+1$
- Angle-refinement from level m to $m+1$
= extension of coefficient ring + lifting of homology class
- Conventional criteria (divisibility, tri-orthogonality, etc.)
 $\Rightarrow$ a part of logicality condition for uniform rotation angle

Relation to high-energy physics?

- Transversal Pauli Z rotation gate at level m
= “ 2^{m-1} -root of Wilson loop”
- Implementability of logical transversal gate
= existence of “gauge-invariant 2^{m-1} -root of Wilson loop”
- Example: $m = 2$

$$W(\gamma) = \exp\left(\oint_{\gamma} a\right) \leftrightarrow Z(\gamma) = \prod_i (Z_i)^{\gamma_i}$$

$$W^{(2)}(\gamma^{(2)}) = \exp\left(\frac{1}{2} \oint_{\gamma^{(2)}} \tilde{a}\right) \leftrightarrow S(\gamma^{(2)}) = \prod_{i=1}^n (S_i)^{\gamma_i^{(2)}}$$

with $W(\gamma) = \left(W^{(2)}(\gamma^{(2)})\right)^2$

$\Rightarrow$ lift of $\mathbb{Z}_2$ gauge field $a(\gamma)$ to $\mathbb{Z}_4$ gauge field $\tilde{a}(\gamma^{(2)})$

Summary

- Gauge theory, quantum error correction and homology theory have **common mathematical structures**
 - Logical gates in QEC codes can be constructed systematically by gauge fields of lattice gauge theory
 - Implementability of transversal logical gates is governed by homological structure of QEC codes beyond $\mathbb{Z}_2$ coefficient
- Exploring **the relation between them** may offer a new insight for comprehensive understanding of them
- Future direction
 - Fault-tolerance in lattice gauge theory or homology theory
 - Non-Abelian extension (non-Abelian codes and quantum double model)
 - Higher order cohomological invariants and non-Clifford gate
 - High-energy physics sides: TQFT interpretation?

Thank you!

Back up

About me

- Name: Junichi Haruna (春名純一)
- Affiliation: Graduate School of Informatics, Kyoto University
- Career:
 - 2018~2023 Ph.D. @ Kyoto University
(particle physics: functional renormalization group)
 - 2023~2026 Postdoc @ IQIB, University of Osaka
 - 2026~present Postdoc @ Kyoto University
(quantum error correction: hierarchical code, gauge field formalism)
- Research interests:
 - Mathematical structure of QEC codes
 - Relation between high-energy physics and QEC

Non-Abelian code and quantum double model

- QEC codes with non-Abelian gauge groups realize non-Abelian anyon and is a key to topological quantum computation
- Quantum double model (Kitaev 1997) :
 - a lattice model with the quantum double of a finite Group G
 - $G = \mathbb{Z}_2$: reduces to the toric code
 - G is non-Abelian : realizes non-Abelian anyon (e.g. $G = S_3, D_4$)
- Non-Abelian anyons : exchange statistics are represented by a matrix
⇒ Universal computation by topological quantum computation!
- Non-Abelian CSS construction?
Ordinary CSS correspondence is on $\mathbb{Z}_2 \rightarrow$ generalization is a hot topic